

CAB112:INTRODUCTION TO DATA SCIENCE

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: gain proficiency in Python programming relevant to data science.

CO2 :: perform EDA techniques to understand data distributions, relationships, and patterns among problems and solutions.

CO3 :: execute machine learning algorithms for regression, classification, and predictive modeling.

CO4 :: develop a comprehensive data science project from start to finish.

Unit I

Introduction to Data Science and Python for Data Analysis: Foundations of Data Science :

Overview of data science and its importance, The data science process and lifecycle

Python Libraries for Data Science : NumPy for numerical computations, Pandas for data manipulation and analysis, Matplotlib and Seaborn for data visualization

Data Collection and Cleaning : Data collection techniques Handling missing data and outliers, Data transformation and normalization

Unit II

Exploratory Data Analysis and Data Visualization: Exploratory Data Analysis (EDA) :

Descriptive statistics Data visualization techniques, Correlation and covariance

Advanced Data Visualization : Advanced plotting with Matplotlib and Seaborn, Interactive visualizations with Plotly and Dash

Unit III

Supervised Learning Algorithms: Regression Algorithms : Linear regression, Polynomial regression, Ridge and Lasso regression

Classification Algorithms : Logistic regression, k-Nearest Neighbors (k-NN), Decision Trees and Random Forests, Support Vector Machines (SVM)

Model Evaluation : Metrics for regression: MAE, MSE, RMSE, R^2 , Metrics for classification: accuracy, precision, recall, F1-score, ROC-AUC, Cross-validation techniques

Unit IV

Unsupervised Learning Algorithms: Clustering Algorithms : k-Means clustering, Hierarchical clustering, DBSCAN

Dimensionality Reduction : Principal Component Analysis (PCA) , t-Distributed Stochastic Neighbor Embedding (t-SNE), Apriori algorithm, Eclat algorithm

Unit V

Advanced Topics in Machine Learning: Neural Networks and Deep Learning : Introduction to neural networks, Deep learning basics with TensorFlow and Keras, Convolutional Neural Networks (CNNs) ? Recurrent Neural Networks (RNNs)

Unit VI

Natural Language Processing : Text preprocessing and normalization, Tokenization and stemming/lemmatization, Bag-of-Words and TF-IDF, Sentiment analysis and text classification

List of Practicals / Experiments:

Practical's

- Perform complex data manipulations such as merging, joining, and group operations on a real-world dataset.
- Implement matrix operations and linear algebraic computations.
- Visualize data distributions using histograms, box plots, and scatter plots.
- Generate pair plots and heatmaps for correlation analysis.
- Normalize features using Min-Max scaling and standardize features using Z-score scaling.
- Use advanced imputation techniques (e.g., KNN imputation) and evaluate their impact on the dataset.

- Generate new features using polynomial transformations, interaction terms, and domain-specific knowledge.
- Build and evaluate a decision tree classifier for predicting fraudulent transactions.
- Implement logistic regression, SVM, and k-NN classifiers on different datasets.
- Implement linear regression, polynomial regression, and support vector regression (SVR).
- Evaluate regression models using metrics such as MAE, MSE, and RMSE.
- Implement Random Forest, AdaBoost, and Gradient Boosting models on a classification problem.
- Apply k-means and hierarchical clustering to segment customers.
- Build and evaluate a convolutional neural network (CNN) for image classification.
- Implement sentiment analysis on social media data using machine learning algorithms.

Text Books:

1. MACHINE LEARNING USING PYTHON by MANARANJAN PRADHAN ,U DINESH KUMAR, NA
2. INTRODUCTION TO DATA SCIENCE:PRACTICAL APPROACH WITH R AND PYTHON by B. UMA MAHESWARI, R. SUJATHA, NA

References:

1. DATA SCIENCE AND MACHINE LEARNING USING PYTHON by REEMA THAREJA, MC GRAW HILL